Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	**TE**	**Semester:**	**V**
Course Code:	**CSC502**	**Course Name:**	**WC**

Name of Student:	**Pranita Kumbhar**
Roll No. :	**70**
Experiment No.:	**09**
Title of the Experiment:	**To configure and verify Static NAT translation**
Date of Performance:	**24/09/25**
Date of Submission:	**04/10/25**

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Ms. Kshitija Gharat

Signature :

Date:

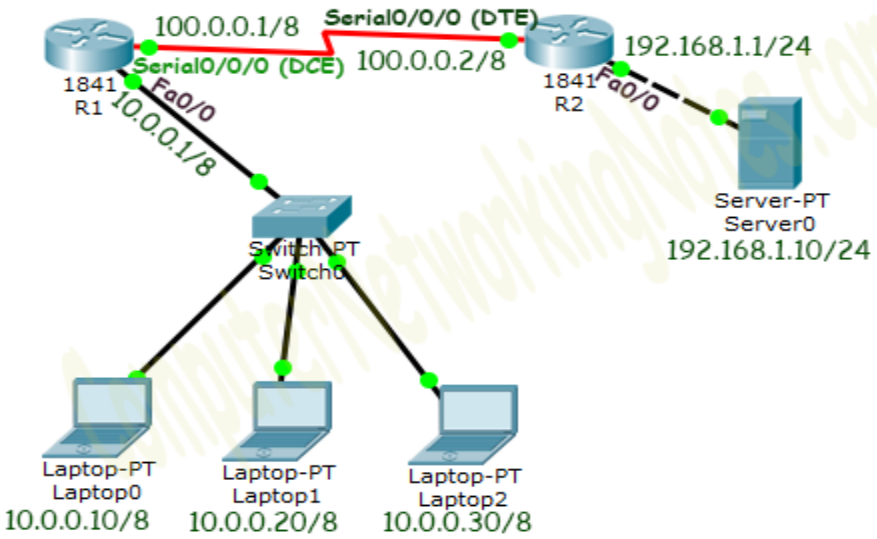
Aim - To configure and verify Static NAT translation

Objective -

1. To create a network topology to understand static NAT configuration
2. To configure static NAT by defining IP addressing, local interface and global interface

Network address translation (NAT) is a method of mapping an IP address space into another by modifying network address information in the IP header of packets while they are in transit across a traffic routing device. The technique was originally used to bypass the need to assign a new address to every host when a network was moved, or when the upstream Internet service provider was replaced, but could not route the network's address space.

Creart a network topology as shown below in Cisco packet tracer



Initial IP Configuration

Device / Interface	IP Address	Connected With
Laotop0	10.0.0.10/8	Fa0/0 of R0
Laptop1	10.0.0.20/8	Fa0/0 of R0
Laptop2	10.0.0.30/8	Fa0/0 of R0
Server0	192.168.1.10/24	Fa0/0 of R1

Serial 0/0/0 of R1	100.0.0.1/8	Serial 0/0/0 of R2
Serial 0/0/0 of R2	100.0.0.2/8	Serial 0/0/0 of R2

Two interfaces of Router1 are used in topology; FastEthernet0/0 and Serial 0/0/0.
Interface mode is used to assign the IP address and other parameters. Interface mode can be accessed from global configuration mode. Following commands are used to access the global configuration mode.

```
Router>enable
Router# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
```

Before we configure IP address in interfaces let's assign a unique descriptive name to router.

```
Router(config)#hostname R1
R1#
```

Now execute the following commands to set IP address in FastEthernet 0/0 interface.

```
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#exit
```

interface FastEthernet 0/0 command is used to enter in interface mode.
ip address 10.0.0.1 255.0.0.0 command assigns IP address to interface.
no shutdown command is used to bring the interface up.
exit command is used to return in global configuration mode.
Serial interface needs two additional parameters clock rate and bandwidth. Every serial cable has two ends DTE and DCE. These parameters are always configured at DCE end.
show controllers interface command from privilege mode to check the cable's end.

```
R1(config)#exit
R1#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
```

DCE V.35, clock rate 2000000

[Output omitted]

Fourth line of output confirms that DCE end of serial cable is attached. If you see DTE here instead of DCE skip these parameters.

Now we have necessary information let's assign IP address to serial interface.

R1#configure terminal

R1(config)#interface Serial0/0/0

R1(config-if)#ip address 100.0.0.1 255.0.0.0

R1(config-if)#clock rate 64000

R1(config-if)#bandwidth 64

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#

Router#configure terminal Command is used to enter in global configuration mode.

Router(config)#interface serial 0/0/0 Command is used to enter in interface mode.

Router(config-if)#ip address 100.0.0.1 255.0.0.0 Command assigns IP address to interface.

Router(config-if)#clock rate 64000

In real life environment this parameter controls the data flow between serial links and need to be set at service provider's end. In lab environment we need not to worry about this value. We can use any valid rate here.

Router(config-if)#bandwidth 64

Bandwidth works as an influencer. It is used to influence the metric calculation of EIGRP or any other routing protocol which uses bandwidth parameter in route selection process.

Router(config-if)#no shutdown Command brings interface up.

Router(config-if)#exit Command is used to return in global configuration mode.

Use commands to assign IP addresses on interfaces of Router2. Clock rate and bandwidth only on DCE side of serial interface need to be provided. Following command will assign IP addresses on interface of Router2.

Initial IP configuration in R2

Router>enable

Router#configure terminal

Router(config)#hostname R2

R2(config)#interface FastEthernet0/0

R2(config-if)#ip address 192.168.1.1 255.255.255.0

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#interface Serial0/0/0

R2(config-if)#ip address 100.0.0.2 255.0.0.0

R2(config-if)#no shutdown

R2(config-if)#exit

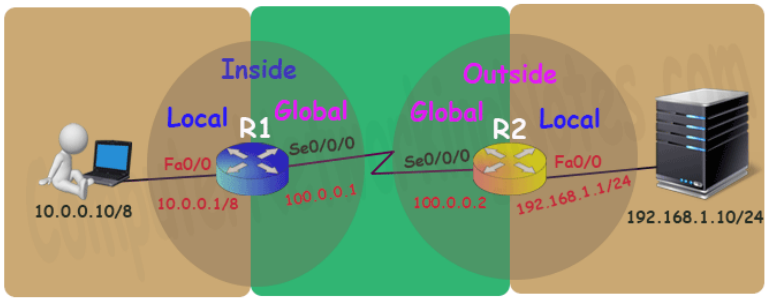
R2(config)#

Configure Static NAT

Static NAT configuration requires three steps: -

1. Define IP address mapping
2. Define inside local interface
3. Define inside global interface

Following figure illustrates these terms.



R1 Static NAT Configuration

R1(config)#ip nat inside source static 10.0.0.10 50.0.0.10

R1(config)#interface FastEthernet 0/0

R1(config-if)#ip nat inside

R1(config-if)#exit

R1(config)#

R1(config)#interface Serial 0/0/0

R1(config-if)#ip nat outside

R1(config-if)#exit

For testing purpose I configured only one static translation. You may use following commands to configure the translation for remaining address.

R1(config)#ip nat inside source static 10.0.0.20 50.0.0.20

R1(config)#ip nat inside source static 10.0.0.30 50.0.0.30

R2 Static NAT Configuration

R2(config)#ip nat inside source static 192.168.1.10 200.0.0.10

R2(config)#interface FastEthernet 0/0

R2(config-if)#ip nat inside

R2(config-if)#exit

R2(config)#

R2(config)#interface Serial 0/0/0

R2(config-if)#ip nat outside

R2(config-if)#exit

Configure static routing in R1

R1(config)#ip route 200.0.0.0 255.255.255.0 100.0.0.2

Configure static routing in R2

R2(config)#ip route 50.0.0.0 255.0.0.0 100.0.0.1

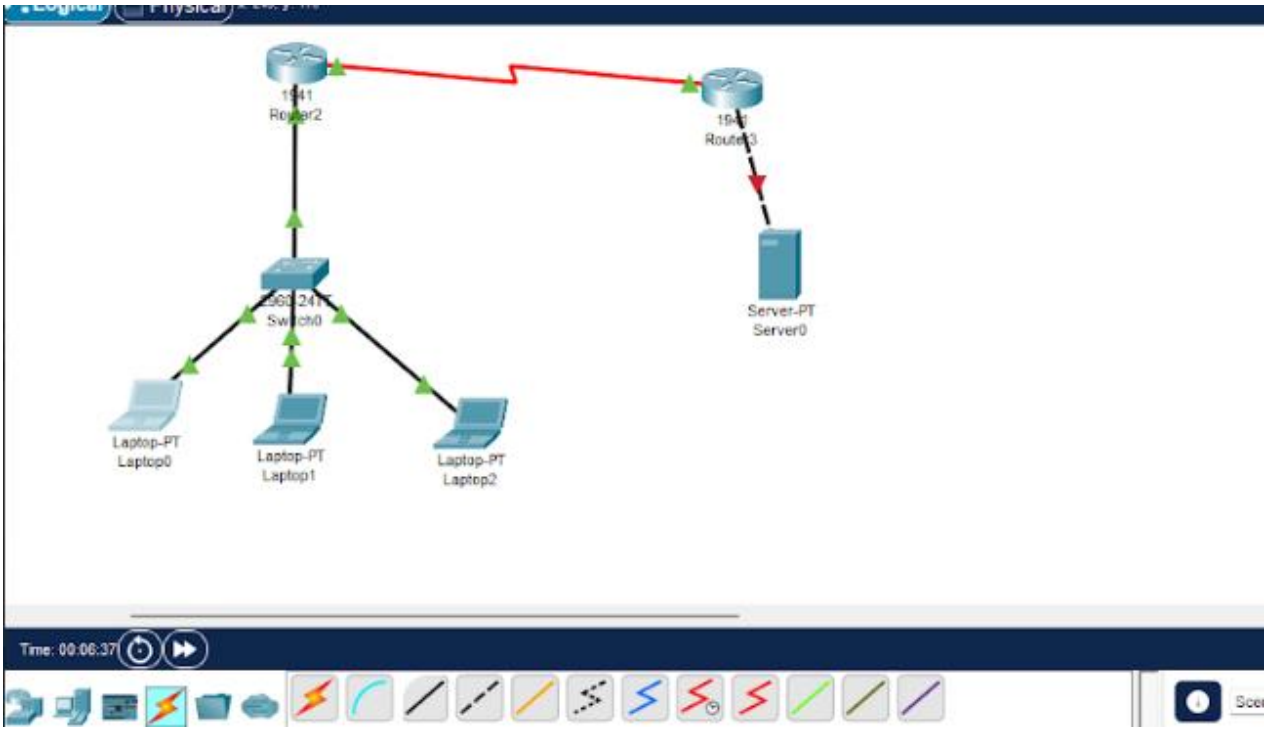
Testing Static NAT Configuration

In this lab we configured static NAT on R1 and R2. On R1 we mapped inside local IP address 10.0.0.10 with inside global address 50.0.0.10 while on R2 we mapped inside local IP address 192.168.1.10 with inside global IP address 200.0.0.10.

Device	Inside Local IP Address	Inside Global IP Address
Laptop0	10.0.0.10	50.0.0.10
Server	192.168.1.10	200.0.0.10

To test this setup click Laptop0 and Desktop and click Command Prompt.

- Run **ipconfig** command.
- Run **ping 200.0.0.10** command.
- Run **ping 192.168.1.10** command.


Server-PT
Server0
Laptop-PT
Laptop0
Laptop-PT
Laptop1
Laptop-PT
Laptop2
Time: 00:06:37

Output:

Physical | Config | Desktop | Attributes | Software/Services

Command Prompt

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Link-local IPv6 Address.........: FE80::260:5CFF:FE8C:4886
   IP Address......................: 10.0.0.10
   Subnet Mask.....................: 255.0.0.0
   Default Gateway.................: 10.0.0.1

C:\>ping 200.0.0.10
Pinging 200.0.0.10 with 32 bytes of data:

Reply from 200.0.0.10: bytes=32 time=13ms TTL=126
Reply from 200.0.0.10: bytes=32 time=14ms TTL=126
Reply from 200.0.0.10: bytes=32 time=13ms TTL=126
Reply from 200.0.0.10: bytes=32 time=12ms TTL=126

Ping statistics for 200.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 14ms, Average = 13ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Conclusion -

Network Address Translation is configured on routers in a network and when it is enabled the IP address of the actual end device could not be traced.